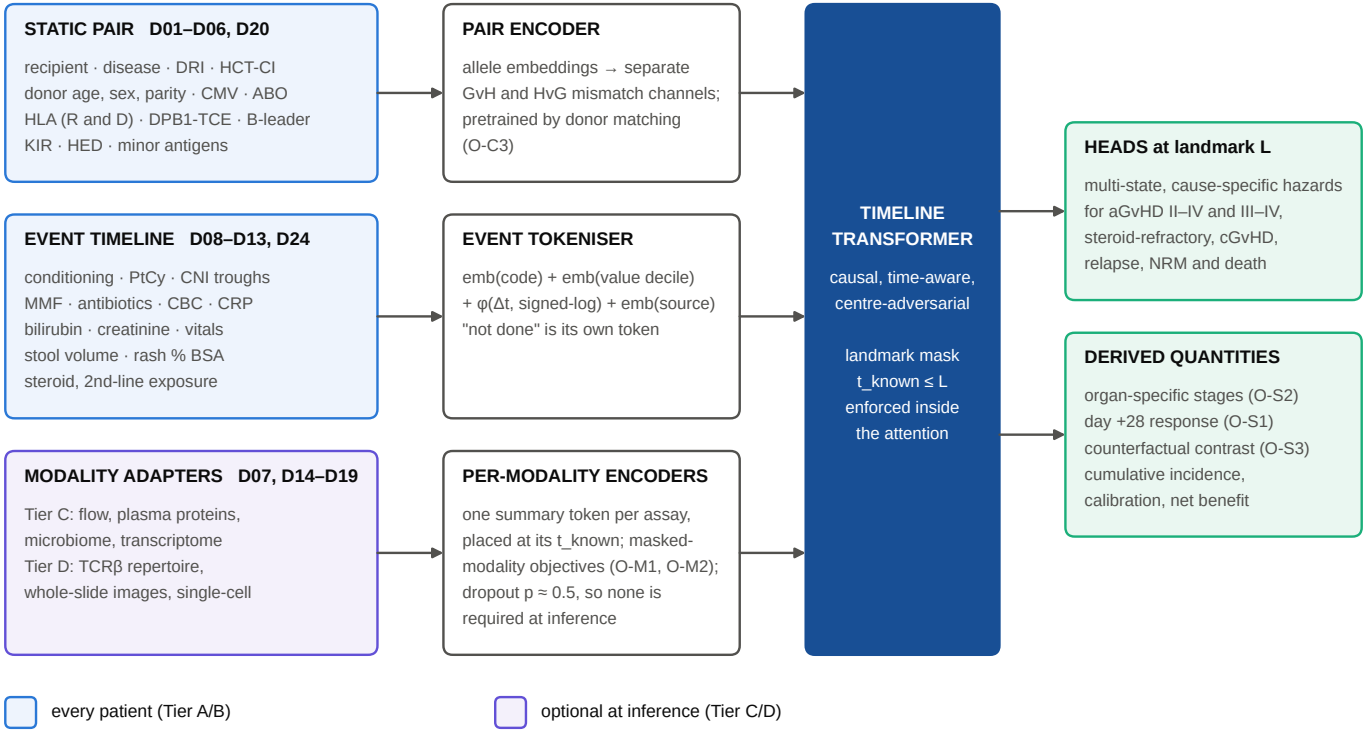
STATIC PAIR D01–D06, D20
recipient · disease · DRI · HCT-CI
donor age, sex, parity · CMV · ABO
HLA (R and D) · DPB1-TCE · B-leader
KIR · HED · minor antigens
EVENT TIMELINE D08–D13, D24
conditioning · PtCy · CNI troughs
MMF · antibiotics · CBC · CRP
bilirubin · creatinine · vitals
stool volume · rash % BSA
steroid, 2nd-line exposure
MODALITY ADAPTERS D07, D14–D19
Tier C: flow, plasma proteins,
microbiome, transcriptome
Tier D: TCRβ repertoire,
whole-slide images, single-cell
PAIR ENCODER
allele embeddings → separate
GvH and HvG mismatch channels;
pretrained by donor matching
(O-C3)
EVENT TOKENISER
emb(code) + emb(value decile)
+ φ(Δt, signed-log) + emb(source)
"not done" is its own token
PER-MODALITY ENCODERS
one summary token per assay,
placed at its t_known; masked-
modality objectives (O-M1, O-M2);
dropout p ≈ 0.5, so none is
required at inference
TIMELINE TRANSFORMER
causal, time-aware,
centre-adversarial
landmark mask
t_known ≤ L
enforced inside
the attention
HEADS at landmark L
multi-state, cause-specific hazards
for aGvHD II–IV and III–IV,
steroid-refractory, cGvHD,
relapse, NRM and death
DERIVED QUANTITIES
organ-specific stages (O-S2)
day +28 response (O-S1)
counterfactual contrast (O-S3)
cumulative incidence,
calibration, net benefit
every patient (Tier A/B)
optional at inference (Tier C/D)